

Pump HOA, Automatic Cycle, and Protection

Complete RSLogix 500 design for Project 10

30-second ON / 10-second OFF cycle, momentary HAND control, flow-loss trip, high-pressure trip, reset, and silence.

Design decision

This solution takes the safer option allowed by the specification: **HAND is inhibited while a fault is active**. HAND remains momentary and never changes the stored AUTO/OFF selection. Therefore, releasing HAND returns the controller to the mode that existed before HAND was pressed.

1 Requirements and I/O

The pump has three logical modes. OFF is the startup state; HAND energizes the pump only while the pushbutton is held; AUTO repeats 30 seconds ON and 10 seconds OFF. A loss of flow for five seconds while the pump is commanded on, or pressure above 30 psi for five seconds, seals in a fault and removes AUTO.

Address	Assigned function	Address	Assigned function
I:0/0	Flow switch	O:0/0	Pump starter
I:0/1	Alarm reset pushbutton	O:0/1	Flow-fault indication
I:0/2	Alarm silence pushbutton	O:0/2	Pressure-fault indication
B3:0/0	HAND pushbutton	N7:0	Raw pressure input
B3:0/1	OFF pushbutton	N7:1	Scaled pressure, psi
B3:0/2	AUTO pushbutton		

1.1 Internal memory

Address	Purpose
B3:1/0	AUTO mode memory
B3:1/1	Spare (AUTO phase is derived from timer status)
B3:1/2	Sealed-in flow fault
B3:1/3	Sealed-in pressure fault
B3:1/4	Flow alert memory (silenceable indication)
B3:1/5	Any-fault summary
B3:1/6	Pressure alert memory (silenceable indication)
B3:2/0, B3:2/2	OSR storage bits for flow and pressure
B3:2/1, B3:2/3	One-scan flow and pressure trigger bits
T4:0	AUTO ON timer, 30.00 s
T4:1	AUTO OFF timer, 10.00 s
T4:2	No-flow timer, 5.00 s
T4:3	High-pressure timer, 5.00 s

2 Pressure Scaling

The pressure transmitter spans 0–55 psi over raw counts 0–16383. Use an **SCP** instruction:

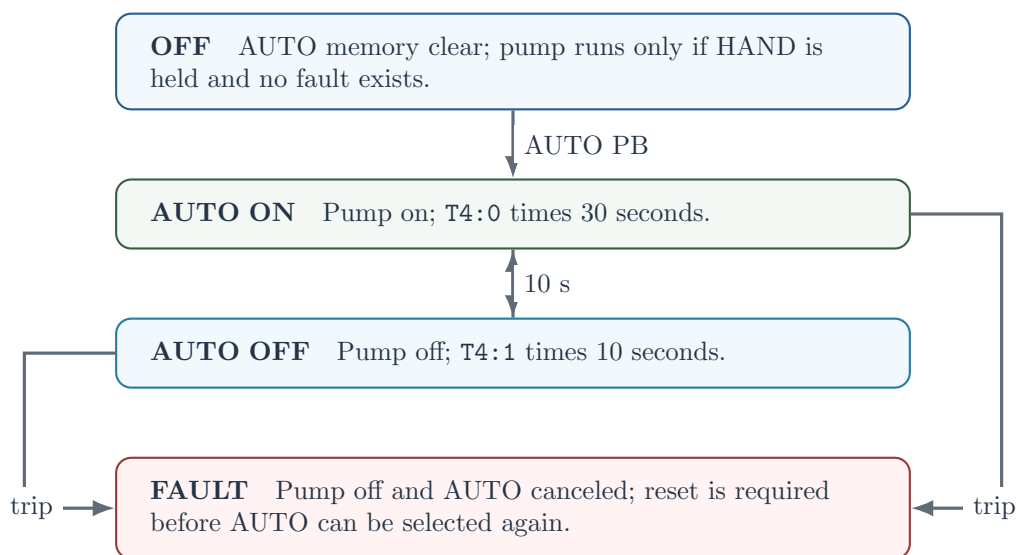
$$P = 55 \frac{N7:0}{16383}.$$

Thus $N7:0 = 8192$ gives $P \approx 27.50$ psi, and $N7:0 = 10000$ gives

$$P = 55 \frac{10000}{16383} \approx 33.57 \text{ psi}.$$

With integer destination $N7:1$, displayed values are rounded to integer psi. Use $F8:0$ instead if fractional display is required; the project tests remain valid either way.

3 Control Structure



HAND is a parallel, momentary demand at the final output rung. It does not alter the sealed-in **AUTO** selection, so it cannot destroy the remembered **AUTO** mode.

4 Complete Ladder Implementation

The following mnemonic rungs are intended as an implementation blueprint. Each timer instruction explicitly shows its time base. With a 0.01-second time base, presets 3000, 1000, and 500 represent 30, 10, and 5 seconds respectively. Branches shown in braces are parallel branches. All stored states use ordinary OTE coils with seal-in contacts; no latch or unlatch instructions are used.

Practical alarm pattern adopted from the sample program

The attached plant program separates each event into a **delay timer**, a one-scan **trigger**, a persistent protective **alarm bit**, and a separately persistent but silenceable **alert bit**. This solution adopts that pattern. **RESET** clears the alarm and alert and resets the delay timer; if the abnormal condition remains, the delay must expire again before a new trip. **SILENCE** clears only the alert/indicator. A later trigger re-annunciates the alarm without disturbing the protective trip state.

Rung 000 – Scale pressure

```
SCP Source N7:0 InMin 0 InMax 16383
  ScaledMin 0 ScaledMax 55 Dest N7:1
```

Rung 001 – Five seconds above 30 psi

```
GRT N7:1 30 XIO I:0/1
  TON T4:3 Time Base 0.01 PRE 500 ACC 0
```

Rung 002 – Create a one-scan pressure trigger

```
XIC T4:3/DN OSR B3:2/2 OTE B3:2/3
```

Rungs 003–004 – Pressure alarm and silenceable alert

The alarm bit is the protective memory. The alert bit drives annunciation and can be silenced independently. Both are set by the same one-scan trigger.

```
{ XIC B3:2/3 } OTE B3:1/3
{ XIC B3:1/3 XIO I:0/1 }

{ XIC B3:2/3 } OTE B3:1/6
{ XIC B3:1/6 XIO I:0/1 XIO I:0/2 }
```

Rung 005 – Fault summary

```
{ XIC B3:1/2 } OTE B3:1/5
{ XIC B3:1/3 }
```

Rung 006 – HOA mode selection

AUTO is accepted only with no fault. Its contact seals around the AUTO pushbutton. OFF or a fault opens the permissive and drops the ordinary output coil, so an operator must deliberately reselect AUTO after reset.

```
XIO B3:0/1 XIO B3:1/5 { XIC B3:0/2 } OTE B3:1/0
  { XIC B3:1/0 }
```

Rungs 007–008 – Automatic 30/10 cycle

The timer done bits define the phase directly. When the OFF timer finishes, it opens the ON-timer rung; both timers reset in sequence and a new ON period begins.

```
XIC B3:1/0 XIO T4:1/DN TON T4:0 Time Base 0.01 PRE 3000 ACC 0
XIC B3:1/0 XIC T4:0/DN TON T4:1 Time Base 0.01 PRE 1000 ACC 0
```

Rung 009 – Final pump command

The outer normally-closed fault contact makes both HAND and AUTO fail safe. HAND is momentary; the AUTO branch uses the stored mode and current phase.

```
XIO B3:1/5 { XIC B3:0/0 } OTE 0:0/0
           { XIC B3:1/0 XIO T4:0/DN }
```

Rung 010 – Five seconds of commanded pumping without flow

Timing from the actual output means the protection works in both HAND and AUTO. If flow arrives before five seconds, the non-retentive timer resets.

```
XIC 0:0/0 XIO I:0/0 XIO I:0/1
           TON T4:2 Time Base 0.01 PRE 500 ACC 0
```

Rung 011 – Create a one-scan flow trigger

```
XIC T4:2/DN OSR B3:2/0 OTE B3:2/1
```

Rungs 012–013 – Flow alarm and silenceable alert

```
{ XIC B3:2/1 } OTE B3:1/2
{ XIC B3:1/2 XIO I:0/1 }

{ XIC B3:2/1 } OTE B3:1/4
{ XIC B3:1/4 XIO I:0/1 XIO I:0/2 }
```

Rungs 014–015 – Alarm indications

```
XIC B3:1/4 OTE 0:0/1 ; flow alert
XIC B3:1/6 OTE 0:0/2 ; pressure alert
```

Scan-order note

The flow fault seals in after the pump rung. On the following PLC scan, the fault summary becomes true, the pump output drops, and AUTO is cleared. This delay is one scan, normally only a few milliseconds. If immediate same-scan removal is required, place a direct **XIO T4:2/DN** permissive in the pump rung as well.

5 How the HOA Behavior Works

HAND from OFF: holding B3:0/0 runs the pump; releasing it returns to OFF.

HAND from AUTO: HAND overrides the 10-second rest portion and keeps the pump on. AUTO phase timing continues in the background, so release immediately returns control to the automatic cycle.

OFF: B3:0/1 opens the AUTO seal-in rung. It does not defeat a held HAND pushbutton, which is conventional HOA behavior.

Fault: either fault inhibits the pump and removes AUTO. This implementation also inhibits HAND until reset.

Silence: the alert bits and indicators turn off, but the alarm bits remain sealed in and continue to inhibit operation. A later trigger re-annunciates the corresponding alert.

6 Verification Against the Test Criteria

Test action	Expected result
Startup; set N7:0=8192 Hold HAND with no flow	OFF mode, no faults, pump off; N7:1 approximately 27–28 psi. The pump energizes. After five seconds, B3:1/2 seals in, pump drops out, and 0:0/1 energizes.
Release HAND; press SILENCE	Flow alert and indication de-energize, but the protective flow alarm remains sealed in. Because this design inhibits HAND during a fault, another HAND attempt does not run the pump.
Press AUTO while faulted	AUTO cannot seal in and the pump does not cycle.
Press RESET; turn flow on; press AUTO	RESET clears alarm and alert memory and holds the alarm-delay timers reset. Pump begins with a 30-second ON phase, then a 10-second OFF phase, repeating continuously.
Hold HAND during AUTO	Pump runs continuously. Releasing HAND returns control to the stored AUTO cycle.
Set N7:0=10000	Scaled pressure is approximately 33.5 psi. After five seconds above 30 psi, B3:1/3 seals in, AUTO clears, pump stops, and 0:0/2 energizes.

7 Implementation Checks

Before testing in Emulate, confirm the following:

1. Each internal alarm or alert bit has exactly one OTE; its own contact provides the seal-in branch.
2. Each OSR uses a unique storage bit, and each trigger bit is written by only one OTE.
3. Every timer is configured for the explicitly shown 0.01-second time base. If a 1.0-second base is selected instead, use presets 30, 10, and 5.
4. The flow-switch polarity is correct. This design assumes I:0/0=1 means flow is proven.
5. The pressure comparison uses the scaled value N7:1, not raw N7:0.
6. The pump output has only one OTE instruction in the entire program.

Result

The program meets the stated sequence and tests while keeping mode memory, protection, and annunciation separate. That separation is what makes the HAND return behavior and alarm silence behavior predictable.